



DALHOUSIE UNIVERSITY

CSCI 5408: Data Management, Warehousing and Analytics

Assignment: Lab - 02

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1) LeetCode Question: 1978. Employees Whose Manager Left the Company

Problem Statement:

Find the IDs of the employees whose salary is strictly less than \$30000 and whose manager left the company. When a manager leaves the company, their information is deleted from the Employees table, but the reports still have their manager_id set to the manager that left.

Return the result table ordered by employee_id.

Query:

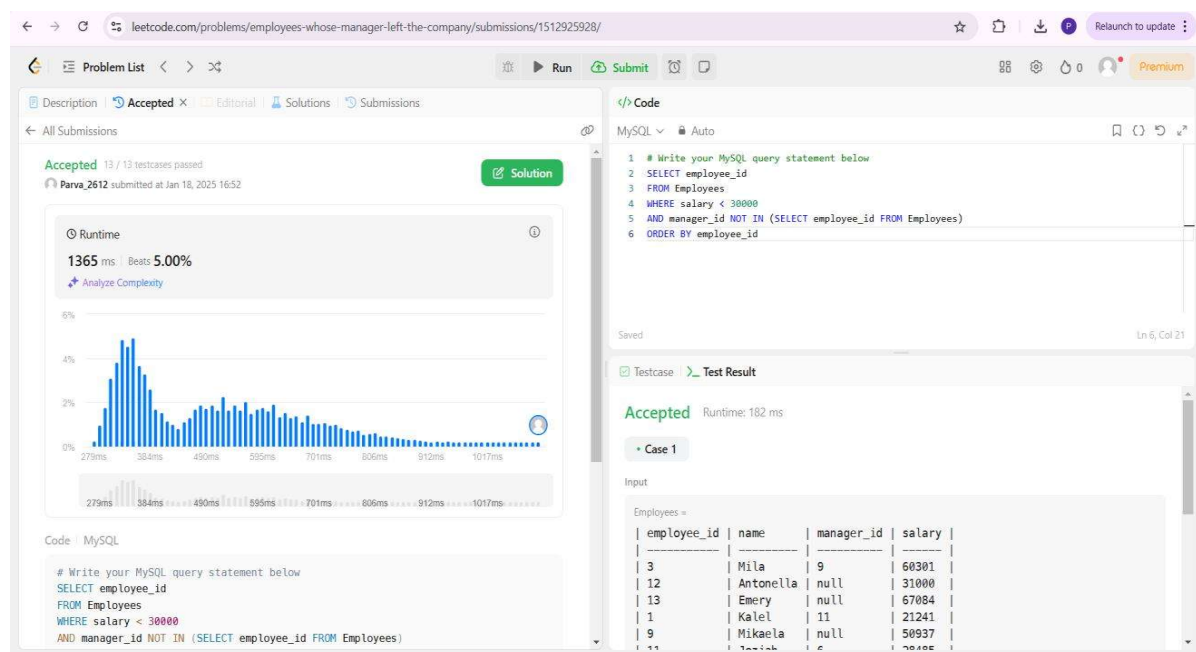
```
SELECT employee_id
FROM Employees
WHERE salary < 30000
AND manager_id NOT IN (SELECT employee_id FROM Employees)
ORDER BY employee_id
```

Explanation:

The following is the explanation of the query

1. The subquery retrieves the **employee_id** values from the **Employee** table.
2. The **where** clause fetches the records of employees whose salary is below 30,000.
3. It also ensures that the **manager_id** is not in the sub list of **employee_id** retrieved by the subquery.
4. Finally, the results are sorted by **employee_id**.

Screenshot:



2) LeetCode Question: 176. Second Highest Salary

Problem Statement

Write a solution to find the second highest distinct salary from the Employee table. If there is no second highest salary, return null

Query

```
SELECT  
(SELECT salary from employee WHERE salary IN  
(select max(salary) from employee  
WHERE salary < (SELECT max(salary) FROM employee)) LIMIT 1)  
AS SecondHighestSalary;
```

Explanation

The following is the explanation of the query

1. The innermost subquery retrieves the maximum salary (**max(salary)**) from the **employee** table.
2. The middle subquery fetches the salaries that are less than the maximum salary retrieved by the first subquery.
3. The outer subquery selects the salary from the **employee** table that matches the maximum salary from the second subquery.
4. The **LIMIT 1** ensures only the one result is returned, even if there are multiple employees with the same second-highest salary.
5. The final result is labelled as **SecondHighestSalary**.

Screenshot

The screenshot displays the LeetCode submission page for the problem "Second Highest Salary". The page includes a runtime graph showing the submission's performance relative to other solutions, with a runtime of 447 ms and a percentile of 41.18%. The submitted SQL query is shown in the code editor, and the test results section indicates that the solution is accepted with a runtime of 142 ms. The input data for the test case is as follows:

id	salary
1	100
2	200
3	300

Reference:

- 1) “SQL LIMIT Clause”, <https://www.geeksforgeeks.org/> [online]. December 05,2024.
Available: <https://www.geeksforgeeks.org/sql-limit-clause/> [Accessed: January 17,2025]

3) Leetcode Questions: 1683. Invalid Tweets

Problem Statement

Write a solution to find the IDs of the invalid tweets. The tweet is invalid if the number of characters used in the content of the tweet is strictly greater than 15.

Return the result table in any order.

Query

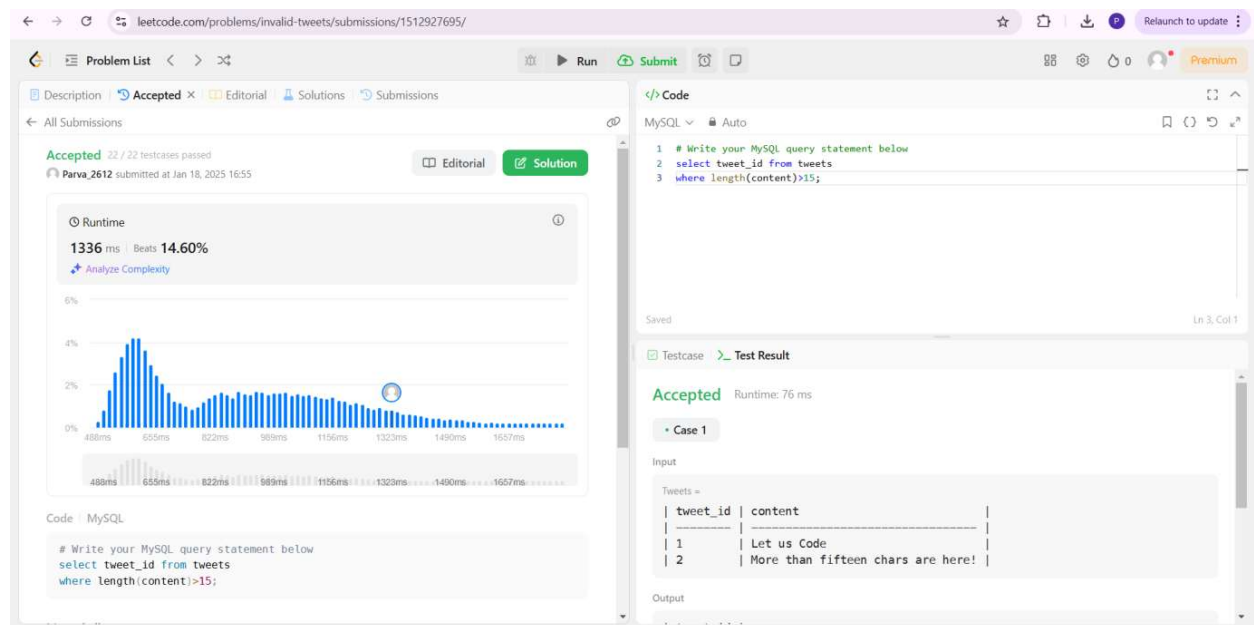
```
SELECT tweet_id FROM tweets
WHERE length(content)>15;
```

Explanation

The following is the explanation of the query

1. The query selects the **tweet_id** from the **tweets** table.
2. It checks the length of each tweet's **content** to see if it is more than 15 characters.
3. The **length(content)>15** condition ensures only tweets with content longer than 15 characters are included.
4. Tweets with 15 characters or fewer are skipped.
5. The query returns the IDs (**tweet_id**) of tweets that meet this condition.

Screenshot



Reference:

- 1) “MYSQL Length() Function”, <https://www.w3schools.com/> [online]. Available:
https://www.w3schools.com/sql/func_mysql_length.asp [Access: January 17, 2025]

4) Leetcode Question: 570. Managers with at least 5 Direct Reports

Problem Statement

Write a solution to find managers with at least five direct reports.

Return the result table in any order.

Query

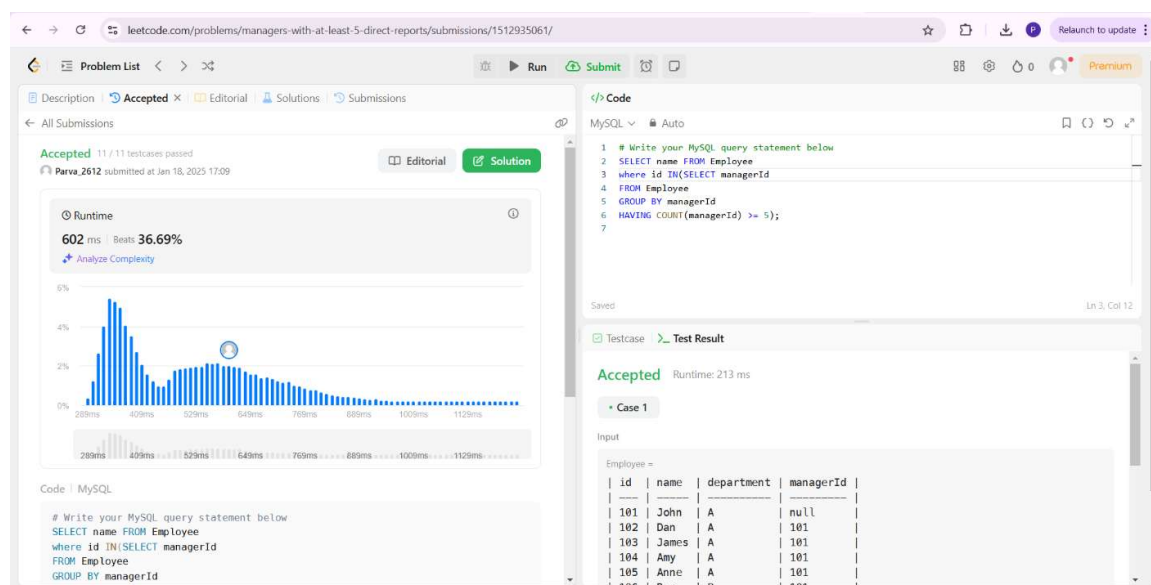
```
SELECT name FROM Employee
WHERE id IN(SELECT managerId
FROM Employee
GROUP BY managerId
HAVING COUNT(managerId) >= 5);
```

Explanation

The following is the explanation of the query

1. The query selects the **name** of employees from the **Employee** table.
2. It focuses on employees whose id matches the **managerId** values in the subquery.
3. The subquery retrieves **managerId** values from the Employee table, grouping them to count how many employees each manager supervises.
4. The **HAVING COUNT (managerId)>=5** condition ensures that only managers with five or more employees are included.
5. The query returns the names of employees who are managers with at least five direct reports.

Screenshot



5) Leetcode Question: 1327. List the Products Ordered in a Period

Problem Statement

Write a solution to get the names of products that have at least 100 units ordered in February 020 and their amount. Return the result table in any order.

Query

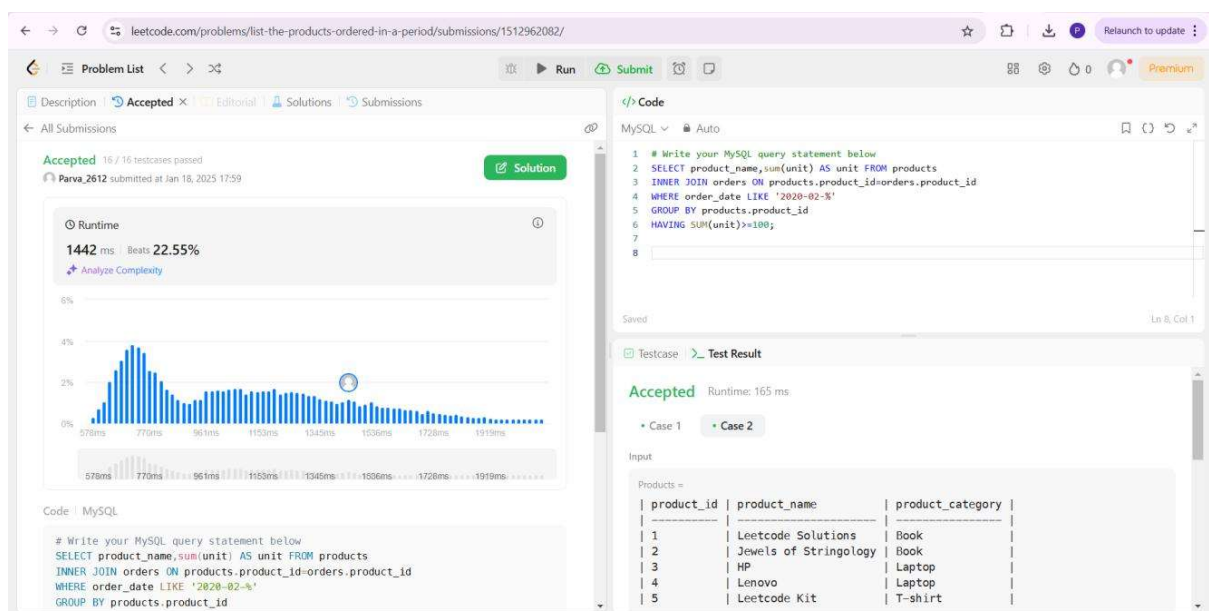
```
SELECT product_name,sum(unit) AS unit FROM products
INNER JOIN orders ON products.product_id=orders.product_id
WHERE order_date LIKE '2020-02-%'
GROUP BY products.product_id
HAVING SUM(unit)>=100;
```

Explanation

The following is the explanation of the query

1. The query retrieves the **product_name** and the total number of units (**SUM(unit)**) from the **products** table.
2. It joins the **products** table with the **orders** table using an **INNER JOIN** on the matching **product_id** values.
3. The **WHERE** clause filters the results to include only orders placed in February 2020, as indicated by order_date **LIKE** '2020-02-%'.
4. The **GROUP BY** clause groups the results by each product's **product_id**, calculating the total number of units ordered for each product.
5. The **HAVING** clause ensures only products with a total of 100 or more units ordered are included in the results.

Screenshot



Reference:

- 1) “SQL LIKE Keyword”, <https://www.w3schools.com/> [online]. Available:
https://www.w3schools.com/sql/sql_ref_like.asp [Access: January 17, 2025]

6) Leetcode Question: 196. Delete Duplicate Emails

Problem Statement

Write a solution to delete all duplicate emails, keeping only one unique email with the smallest id.

For SQL users, please note that you are supposed to write a DELETE statement and not a SELECT one.

Query

```
DELETE p FROM Person as p
INNER JOIN Person as p1
ON p.email = p1.email
AND p.id > p1.id;
```

Explanation

The following is the explanation of the query

1. The query deletes row from the **Person** table using the alias **p**.
2. It compares the table to itself to identify duplicate rows based on the **email** column.
3. The condition **p.id > p1.id** ensures only the row with the higher **id** value is deleted for each duplicate email.
4. This removes duplicate records, keeping the one with the smallest **id** for each email.

Screenshot

The screenshot displays a LeetCode submission for the problem "Delete Duplicate Emails". The submission is marked as "Accepted" with a runtime of 2004 ms, beating 9.58% of other submissions. The code is a MySQL query that deletes duplicate emails by joining the Person table with itself and deleting rows where the email is the same but the id is greater. The test result shows the input table with three rows, including a duplicate email, and the output table with two rows, where the duplicate has been removed.

```
# Write your MySQL query statement below
DELETE p FROM Person as p
INNER JOIN Person as p1
ON p.email = p1.email
AND p.id > p1.id;
```

id	email
1	john@example.com
2	bob@example.com
3	john@example.com

id	email
1	john@example.com
2	bob@example.com